Fruit juice (not canned) — HMTc certification standard

Institute of Contaminant Standards — Heavy Metal Tested & Certified program

Version 1.0 · Native basis: as_consumed · 7/8 analytes with a published limit

STANDARD

fruit-juice, version 1.0. HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

PROVENANCE

Compiled from data/evidence/regulatory_limits.csv. Each limit is a lookup against sovereign law, not an occurrence computation.

TAMPER-EVIDENT SEAL

The SHA-256 of this exact PDF is sealed by an RFC-3161 trusted timestamp. The token (standard.pdf.tsr) and the manifest (manifest.json) accompany this file; the date of record is the timestamping authority's, not the program's.

Abstract

This is the frozen, version-pinned Heavy Metal Tested & Certified standard for fruit juice (not canned). Each per-analyte certification limit is the strictest heavy-metal maximum level set by any government in the world for this product class, expressed in the product's native basis (as_consumed) and derived by lookup against sovereign law under HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The per-jurisdiction derivation below is the complete "show your work". This snapshot is immutable: any material change to a limit mints a new version with its own timestamped snapshot.

1 | Certification limits

All values in ppb (µg/kg), as_consumed basis. A brand meeting every published value on a like-for-like basis qualifies for the mark. Tier 1 toxics (Pb, Cd, iAs, mercury) carry a zero-tolerance posture; Tier 2 metals (Ni, Al, Sn, total Cr) carry the program's transitional allowance. Where a limit is set in another basis by the binding jurisdiction it is converted into the native basis and the conversion is named in the "Basis derivation" column.

both

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	20	European Union	face value	regulatory-alignment
Inorganic arsenic (iAs)	Tier 1	10	US-FDA	face value	regulatory-alignment
Cadmium (Cd)	Tier 1	10	Israel	face value	regulatory-alignment
Mercury (tHg)	Tier 1	10	Israel	face value	regulatory-alignment
Nickel (Ni)	Tier 2	250	European Union	face value	regulatory-alignment

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Aluminium (Al)	Tier 2	500	—	face value (nearest government value; comparator vs any direct limit)	read_across
Tin (Sn)	Tier 2	5000	—	face value (nearest government value; comparator vs any direct limit)	read_across
Total chromium (Cr)	Tier 2	monitored	—	—	not_material

2 | Per-jurisdiction derivation

The binding value for each analyte is the lowest sovereign maximum level across the jurisdictions on record, in the native basis. Rows marked "converted" were set by the jurisdiction in the other basis and carried across the reconstitution ratio; rows marked "native" were already in this basis. Every value is keyed to its record in the sovereign register.

both

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	European Union-(binding)	20	converted	eu2023_pb_infant_drinks_ready_20
—	Codex Alimentarius	30	converted	codex_pb_fruit-juices-whole-commodity-or--reconsti_30
—	Gulf (GSO)	30	converted	gso193_2021_pb_fruit_juices_30
—	HK	30	converted	hk_cap132v_pb_fruit-juices-other-than--fruit-juices-exclusively-from-b_30
—	Israel	30	converted	il_fcs2016_pb_18_2_fruit_and_vegetable_-juices_including_nectars_except_30
—	MA	30	converted	ma_2410_2022_pb_3_1_19_jus_de_fruits_jus_-de_fruits_concentres_reconstitues_et_-nectars_de_frui_30
—	Thailand	30	converted	th_moph414_2020_pb_fruit_juices_general_30
—	United Kingdom	30	converted	uk_pb_drinks-for-infants-and-young--children-la_30
—	Colombia	50	converted	co_res4506_2013_pb_fruit_juices_50
—	India	50	converted	in_fssai2016_pb_037_fruit_juices_-including_nectars_ready_to_drink_50
—	MERCOSUR	50	converted	mercosur_gmc12_2011_nonalcoholic_-beverages_excluding_juices_pb_50
—	PY	50	converted	py_decreto1335_2019_nonalcoholic_-beverages_excluding_juices_pb_50
—	US-FDA	50	converted	fda2004_juice_haccp_pb_50
—	VN	50	converted	vn_qcvn8_2_2011_pb_vegetable-and-fruit--juice-including-nectar-ins_50

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—	South Korea	100	converted	kr_pb_other-foods-for-infants-and-young--childr_100
—	TT	200	converted	tt_fooddrugs_2007_div6_fruit_juice_except_-apple_pb_200
—	Singapore	300	converted	sg_foodregs_2026_fruit_vegetable_juice_-excl_lime_lemon_pb_300
—	UY	300	converted	uy_rbn_pb_citrus_fruit_juices_0_3mgkg
iAs	US-FDA (binding)	10	converted	fda2023_ias_apple_juice_10
—	European Union	20	converted	eu2023_ias_fruit_juice_20
—	Canada	30	converted	ca_ias_grape-juice-and-grape-nectar--general-foo_30
—	Israel	200	converted	il_fcs2016_ias_16_2_fruit_and_vegetable_-juices_including_nectars_200
Cd	Israel (binding)	10	converted	il_fcs2016_cd_17_1_beverages_juices_and_-nectars_10
—	European Union	20	converted	eu2023_cd_infant_drinks_ready_20
—	MERCOSUR	20	converted	mercosur_gmc12_2011_nonalcoholic_-beverages_excluding_juices_cd_20
—	PY	20	converted	py_decreto1335_2019_nonalcoholic_-beverages_excluding_juices_cd_20
tHg	Israel (binding)	10	converted	il_fcs2016_thg_17_1_beverages_juices_and_-nectars_10
Ni	European Union-(binding)	250	converted	eu2024_ni_juice_250
Sn	CL	150000	converted	cl_decreto977_1996_sn_blackcurrant_small_-fruit_nectars_150000
—	UY	150000	converted	uy_rbn_sn_citrus_juices_not_canned_150mgkg

3 | Method and provenance

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The default standard for any (product, analyte) is the strictest maximum level set by a credible government regulator, converted to the row's native basis. Where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. Species limits are bounded by their parent total (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ total Cr); read-across limits (for example aluminium carried from an as-consumed limit to an as-sold basis across the reconstitution ratio) are disclosed as read-across and never presented as direct governance.

The occurrence and toxicology corpus that contextualises these limits is maintained independently as the Heavy Metal Index literature baseline (heavymetalindex.com). This certification standard cites that baseline one way; it does not use the baseline's contents as the basis for any government-limit value stated here.

4 | Version and tamper-evidence

This is version 1.0 of the Fruit juice (not canned) standard. A certification number becomes final when its version carries a resolvable DataCite DOI; a material change to any limit mints a new version with its own frozen snapshot, so a superseded number is never silently overwritten. Tamper-evidence does not depend on trusting the program: the SHA-256 of this exact PDF was submitted to an RFC-3161 trusted timestamping authority (https://freetsa.org/tsr), which returned a signed token binding that hash to a time it — not the program — attests. The token (standard.pdf.tsr), the authority's CA chain (tsa-cacert.pem), and the manifest (manifest.json) travel with this file.

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